An attractor account of numerosity perception

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Abstract

Numerical perception is classically split between an exact system for small sets and an approximate one for larger sets. An alternative tradition posits a single, discrete numerosity representation but leaves open how one code could also yield approximate numerosity extraction. We give this latter idea a mechanistic form: integer-centred memory components form an attractor landscape on the number line, with component width increasing with magnitude, and a noisy numerical input initializes the landscape and is stabilized by pattern completion. Approximation at larger numerosities then emerges as the error of a single discrete code rather than a separate, noisy system. Fitted to individual visual-enumeration responses, the model outperforms matched static and constant-width controls, a purely driven accumulator, and Weber-style and bounded-optimal alternatives. Its dynamics improve the response fit and predict response-time structure from bounded convergence time. As component width grows with magnitude, the same landscape shifts gradually from discrete to increasingly approximate readout.

Keywords

numerical cognition, subitizing, exact number system, attractor dynamics, approximate number system

Introduction

Humans, like many non-human animals, possess a non-symbolic ability to extract numerical information from the environment and represent the numerosity of sets, a capacity commonly termed the number sense (Dehaene 2011; Nieder 2016). This ability is typically expressed in two characteristic behavioural regimes: small sets, typically up to three or four items, are apprehended rapidly and with high accuracy, a phenomenon known as subitizing (Kaufman et al. 1949; Mandler and Shebo 1982). Beyond this range, responses become slower and less accurate, response variability grows with set size, and discrimination depends primarily on the ratio between numerosities (Feigenson et al. 2004; Dehaene 2011). The standard two-systems account attributes these regimes to distinct mechanisms: a capacity-limited Object Tracking System (OTS), which individuates a few objects in parallel, and an Approximate Number System (ANS), which represents larger numerosities as noisy analogue magnitudes (Feigenson et al. 2002; Piazza et al. 2011).

An alternative account of the number sense treats both regimes as outputs of a single code for discrete numerosities. Gallistel and Gelman proposed that a non-verbal counting process represents discrete, countable quantities as analogue magnitudes generated by successive increments and arranged on a linear scale (Gallistel and Gelman 1992, 2000). These representations exhibit scalar variability: their standard deviation grows in proportion to magnitude, so neighbouring magnitude distributions overlap increasingly at larger values. This provides a route by which one code can support highly accurate small-number judgements while yielding increasingly graded, ratio-dependent errors for larger numerosities, without assigning the two regimes to separate systems (Whalen et al. 1999; Cordes et al. 2001).

Computational work has developed several single-system explanations of the two behavioural regimes. The central comparison for the present study is the pair of capacity-limited accounts developed by Cheyette, Piantadosi, and colleagues. At a normative level, Cheyette and Piantadosi (2020) ask how a system with finite information-processing capacity should encode numerosities under a power-law prior. The resulting bounded-optimal code allocates greater precision to small values and progressively coarser precision to larger ones, producing near-exact small-number performance and Weber-like variability without a change of numerical system. At a more visually explicit level, Cheyette et al. (2024) locate the relevant bottleneck in limits on visuospatial memory, which can generate the characteristic psychophysics of numerosity from visual arrays. An earlier, representationally distinct model by Zorzi and Butterworth (1999) established a related point: numerical size and distance effects can emerge from a linear cumulative code coupled to nonlinear response competition, without assuming logarithmic compression or scalar representational noise. More recently, Malo et al. (2023) showed that a small quantum-spin model can encode numerosity in the spectrum of its collective dynamics, providing a distinct dynamical route to a unified numerical representation. Other models have derived numerical sensitivity from recurrent visual dynamics (Sengupta et al. 2014, 2017) or learned image-computable representations (Stoianov and Zorzi 2012;

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Testolin et al. 2020). Together, these accounts show that the two behavioural regimes need not imply two numerical systems, while leaving open whether their common source lies in capacity-limited sensory encoding or in the structure and dynamics of a shared numerical representation.

These visual accounts treat numerosity as a product of modality-specific sensory processing, whether as a consequence of limited visuospatial capacity or as a quantity extracted from the image. Number, however, behaves as an abstract quantity that is not bound to any one sense. It transfers across modalities, so that adapting to a stream of sounds or touches shifts the apparent numerosity of a visual array (Arrighi et al. 2014), its acuity tracks numerical experience, sharpening with formal schooling in otherwise comparable adults (Piazza et al. 2013; Nys et al. 2013), and it depends causally on parietal cortex, where stimulating the intraparietal sulcus disrupts numerosity while sparing lower-level vision (Cappelletti et al. 2007; Dormal et al. 2012) and focal damage yields acalculias that extend even to enumerating touches (Ashkenazi et al. 2008; Cohen et al. 2018). This picture is not uniform: nonsymbolic number stays robust with little formal schooling once memory and symbolic demands are minimized (Frank et al. 2008; Zebian and Ansari 2012; Pica et al. 2004), so this dichotomy is still under debate. Together, these findings motivate an architecture in which modality-specific sensory processing feeds a shared numerical representation.

We propose a one-dimensional, Hopfield-inspired energy landscape as this shared numerical stage (Hopfield 1982; Ramsauer et al. 2021). Each integer contributes a stored memory component, and these components jointly generate a sequence of attractor basins that becomes broader and shallower with magnitude. We motivate this landscape geometry by the cross-linguistic power-law decline in number-word frequency (Dehaene and Mehler 1992), so that more frequently encountered numbers form narrow, deep basins, whereas less frequent numbers form broad, shallow ones. A one-dimensional estimate y_0 of numerosity, supplied by modality-specific sensory processing, initializes the landscape, followed by recurrent clean-up that stabilizes this evidence through pattern completion. The attractor therefore models the transformation from noisy numerical evidence to an integer response, while the same dynamics link response time to settling time.

On this view, a single numerical landscape supports two forms of readout. Where neighbouring integer attractors are sufficiently distinct, pattern completion stabilizes noisy evidence around a categorical value, yielding rapid and highly accurate responses. As magnitude increases, adjacent basins broaden, become shallower, and increasingly overlap, so categorical resolution is progressively lost; the same landscape then produces coarser, ratio-dependent estimates. The transition from resolved to approximate responding is therefore not implemented as a switch between numerical systems, but emerges from the geometry of one representation. The model thus gives mechanistic form to the single-code architecture proposed by Gallistel and Gelman, while adding an explicit representational mechanism: integer-centred attractors embedded in a continuous, noisy state space (Gallistel and Gelman 1992, 2000).

We test this account on ∼53,000 individual responses from the brief-exposure dataset of Cheyette and Piantadosi (2020), spanning five presentation durations from 40 to 640 ms. It reproduces both the underestimation of larger numerosities and the fall-off of exact accuracy with set size, and outperforms a Weber/ANS baseline and their own bounded-optimal model. Turning on the clean-up dynamics improves the fit at the cost of a single shared give-up budget, while the attractor's settling time predicts response times, whose plateau at larger numerosities finds no natural account in the model without that budget. At larger numerosities, the fitted response distributions recover Weber's law, with an effective Weber fraction closely matching the data. In an exploratory reanalysis of developmental dyscalculia, the model further links poorer exact enumeration to a coarser width profile rather than increased sensory noise.

Methods

We thus consider the problem of how our model maps a noisy perceptual input onto a discrete numerical response. We posit an energy landscape defined over a continuous axis x, with attractor wells associated with integer positions (Fig. 1a). In the visual task analysed here, the initial state y_0 summarizes evidence extracted from a dot array; formally, the same interface could receive scalar evidence from another sensory modality. Starting from $x = y_0$, the system evolves by gradient descent on the energy landscape until x settles into a stable well, implementing Hopfield-style pattern completion (Hopfield 1982). A probability distribution over integers is then read out from the final resting state x^*. The settling dynamics also determine how long the trial takes.

In our landscape, each integer n acts as a stored memory represented by a normalized Gaussian component centred at n. Its scale σ_n controls both the breadth and the effective depth of the resulting basin: increasing σ_n broadens the component and, through the normalization term $-\log \sigma_n$, lowers its peak contribution to the log-sum-exp landscape. We use the cross-linguistic power-law decline in number-word frequency (Dehaene and Mehler 1992) to motivate this coupled scaling, so frequent small numbers form narrow, deep basins and rarer large numbers form broad, shallow ones. Accordingly, the Gaussian scale grows as a power of magnitude,

$$\sigma_n = \sigma_c \, n^p, \tag{1}$$

where σ_c sets the overall scale and the exponent p controls how fast resolution degrades with magnitude. Each memory component contributes a log-density

$$g_n(x) = -\log \sigma_n - \frac{(x-n)^2}{2\,\sigma_n^2}. \tag{2}$$

Using the log-sum-exp energy form associated with modern Hopfield networks (Ramsauer et al. 2021), the attraction potential is the log-sum-exp over all wells, $V(x) = -\log \sum_n \exp\big(g_n(x)\big)$.

A sensory estimate $y_0 \sim \mathcal{N}\big(n,\ s_{00}^2 + s_0\, n^{\lambda_0}\big)$, whose noise grows with numerosity ($s_{00} = 0.05$ fixed; s_0 and λ_0 fitted), starts the descent on integer wells at $m = 1, \ldots, 19$.

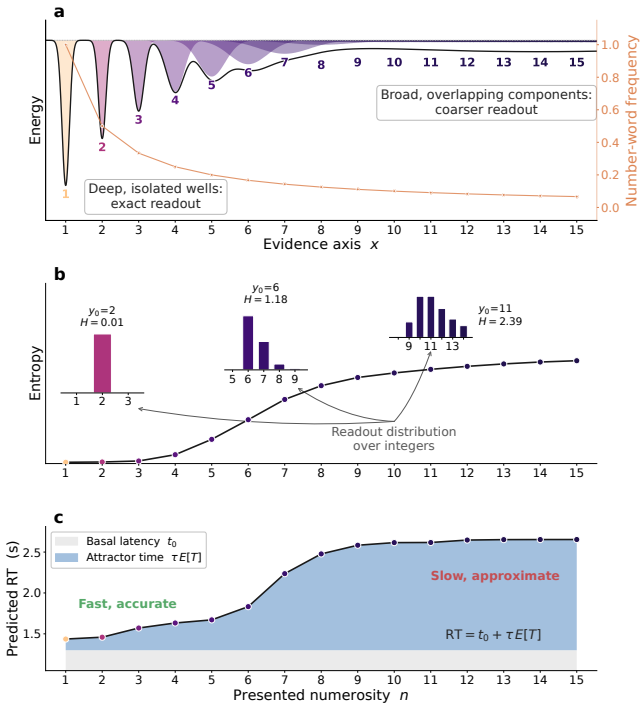


Figure 1. The attractor model of the number line (parameters fitted at 160 ms). a, Energy landscape over the internal evidence axis x: each integer contributes a memory component whose width grows with magnitude (the orange curve is the declining number-word frequency that motivates this widening), so small numbers sit in deep, isolated basins that clean up to an exact value. Figure 2c reports the width index n_{width}. **b**, Expected readout entropy $E[H]$ per presented numerosity, before (grey) and after (black) clean-up, with the attractor's expected convergence time $E[T]$ (blue, right axis); insets show example post-clean-up posteriors. Clean-up removes the residual uncertainty of the small-n readout at the cost of a few settling steps; near the boundary the settling time grows and the residual entropy rises. **c**, The selected response-time link, $\mathrm{RT} = t_0 + \tau\, E[T]$, decomposed into basal latency and convergence time.

From $x = y_0$ the state descends the potential, $x_{t+1} = x_t - \eta\, \partial V/\partial x$ with fixed $\eta = 1$ and a global step clamp (full numerical settings in Supplementary Material 1). Because V is a log-sum-exp over the stored integers, each step moves the state towards a softmax combination of the wells, weighted by their precisions, so every stored integer contributes to the final position. The descent stops when the update falls below a fixed tolerance or reverses sign (oscillation); x^* is then the current state and T the iteration at stopping. On the almost flat, large-numerosity landscape the state does not settle within a single well, so the clean-up never converges and would run indefinitely. Therefore, we introduce a hypothesized give-up mechanism that stops the clean-up process after G steps, giving $T = \min(T^*, G)$, where T^* is the unbounded settling time. We fix G at the profile-likelihood optimum, $G = 120$ (Supplementary Material 1). A distribution over integers is then read out, $p(n \mid x^*) = \mathrm{softmax}\big(g_n(x^*)\big)$, from which we take the mean estimate and the most probable integer.

The model gives a natural interpretation of response time through the time required for the clean-up mechanism to settle on a well:

$$\mathrm{RT} = t_0 + \tau\, \mathbb{E}[T], \tag{3}$$

where T is the number of clean-up steps, bounded by the global budget G, and the expectation is taken over sensory noise. Observed response times are modelled with a log-normal likelihood centred on this value. To test whether static readout uncertainty contributed beyond the attractor dynamics, we compared this convergence-time link with an entropy-only alternative, $t_0 + \kappa\mathbb{E}[H]$, and with the hybrid $t_0 + \tau\mathbb{E}[T] + \kappa\mathbb{E}[H]$. All mapping coefficients were shared across exposure durations, whereas the residual RT variance was duration-specific. The basal latency $t_0 = 1.30$ s is task-fixed: recorded RT starts at stimulus offset, so t_0 absorbs mask, typing, and motor time rather than exposure (Supplementary Material 1).

We fit the model to the individual responses of Cheyette and Piantadosi (2020), in which adult participants briefly viewed dot arrays of 1–15 items and typed an estimate, at five exposure durations (40, 80, 160, 320, 640 ms). The source paper analysed four between-experiment arms at $N = 100$ each after excluding the ten highest-error participants per arm; the open DURATION file used here pools the three duration-control arms without that exclusion and is within-subject across exposures (379 sessions; 360 completed all five durations). We evaluate the categorical likelihood of the 52,995 individual integer reports. For each numerosity the response probability marginalizes the readout over the sensory noise,

$$p(r \mid n) = \int p\big(r \mid x^*(y_0)\big)\, \phi\big(y_0; n, \sigma(n)\big)\, dy_0, \tag{4}$$

with $\sigma(n) = \sqrt{s_{00}^2 + s_0\, n^{\lambda_0}}$, computed by deterministic grid quadrature (Supplementary Material 1). Typed responses run well beyond the well support, so we add a uniform lapse channel, $p_{\text{final}}(r \mid n) = (1 - \ell)\, p(r \mid n) + \ell\, /\, R_{\max}$, with ℓ fitted per duration and counted in all information criteria. In the joint fit the landscape (σ_c, p) and give-up budget G are shared across durations, while (s_0, λ_0, ℓ) are duration-specific ($k = 18$; Supplementary Material 2), fitted by differential evolution (Storn and Price 1997) and Nelder–Mead polishing. With the landscape held fixed, the selected RT channel adds one shared convergence-time coefficient τ and a per-duration residual σ_{rt}. Full numerical settings and fitted values are in Supplementary Material 1.

We fitted the model with a joint fit in which a single power-law landscape (σ_c and p) and the give-up budget G were shared across all five exposure durations, while the sensory channel and lapse were allowed to vary by duration. Models were compared under the same trial-level likelihood against a graded set of alternatives. Matched static readouts isolate the contribution of clean-up for both width laws, whereas constant-width controls fix $p = 0$ to isolate the contribution of magnitude-dependent broadening (as opposed to the effect of increasing noise variance). A Weber/ANS code and the bounded-optimal model of Cheyette and Piantadosi (2020) benchmark the account as a whole. Parameter sharing, total model flexibility, and the corresponding AIC and BIC values are reported in Supplementary Table 2.

To test whether the attractor could itself implement numerical accumulation, we fitted a purely driven variant

with no completed scalar estimate y_0. During exposure, registered items generated a running count $K(t)$, which drove the state while the attractor force acted,

$$\tau_x \dot{x} = -\frac{\partial V(x)}{\partial x} + \beta\big(K(t) - x\big).$$

At stimulus offset the drive was removed, and the response was read from the resulting state. This variant tests whether serial evidence accumulation and associative clean-up can be implemented by the same dynamics.

We summarize the fitted width law with the width index n_{width}, defined by $\sigma_n = 1$, where the fitted Gaussian-component width equals the unit spacing between integers. With $\sigma_n = \sigma_c n^p$, this gives $n_{\text{width}} = \sigma_c^{-1/p}$. The definition is equivalent to the heuristic $d' = \text{spacing}/\sigma_n = 1$ (Macmillan and Creelman 2004) and summarizes the combined effect of the fitted scale and exponent.

Finally, as an exploratory application, we fit the same model to the exact-enumeration data of Decarli et al. (2020) from children with and without developmental dyscalculia, localizing the group difference within the landscape (Supplementary Material 5).

Results

Among the models considered, the power-law attractor provided the best account of the trial-level responses (Fig. 2a; Supplementary Material 2). The constant-width attractor trailed by $\Delta\text{AIC} \approx 3488$, the Weber/ANS code by $\Delta\text{AIC} \approx 838$, and the bounded-optimal model of Cheyette and Piantadosi (2020) by $\Delta\text{AIC} \approx 11{,}453$. We also analyse the effect of the attractor dynamics on model performance. If we switch them off, so that we have a static readout of the same wells with one parameter fewer (it needs no budget), the model fits worse by $\Delta\text{AIC} \approx 722$ under the joint fit. The dynamics therefore sharpen the response distribution at the cost of a single global budget parameter. Finally, the purely driven accumulator fitted substantially worse than the completed-estimate attractor ($\Delta\text{AIC} = 7954$; Fig. 2a). In this purely driven model, as the running count increased, the state was repeatedly drawn towards intermediate integer representations and often had not reached the presented numerosity by stimulus offset (Supplementary Fig. S1).

Because clean-up is a dynamical process, its convergence time provides a direct predictor of response time. The convergence-only and hybrid links were effectively indistinguishable ($\Delta\text{AIC} = 0.24$), so we retained convergence time alone as the more parsimonious model. The entropy-only link performed substantially worse ($\Delta\text{AIC} = 1579.5$; Fig. 2b). Thus, under a single latent-to-time mapping shared across exposure durations, residual readout entropy did not add supported predictive value beyond convergence time. Nevertheless, since clean-up does not converge on the flat large-numerosity landscape, its settling time would grow without bound; the give-up mechanism caps it after G steps, so the predicted response time plateaus onto the data rather than diverging past the budget's onset (Fig. 3c, vertical line).

The fitted landscape has wells with width growing with magnitude, so they are sharp and well separated for small numbers and broad and overlapping for large ones. Since σ_c and p are identifiable under a parameter recovery analysis (Supplementary Material 3), we summarize the fitted width law with the operational index $n_{\text{width}} = \sigma_c^{-1/p}$, which combines its scale and exponent into a single profile-level measure. In the joint fit, the shared width law gives $n_{\text{width}} = 8.5$, which coincides with the onset of the response-time plateau (Fig. 1c).

To further study the individual differences, we compare the landscape resolution and the numerical acuity measured via the Weber ratio. These are, as expected, negatively correlated under empirical-Bayes estimates ($\hat{\rho} = -.57$). The hierarchical correlation between the participant effects on $\log n_{\text{width}}$ and $\log w$ is $\rho = -0.569$ at the primary split $n \le 8$ vs. $n \ge 9$ ($N = 362$; likelihood-ratio $\hat{\rho} = -.569$, likelihood-ratio $\chi^2(1) = 156.9, p < 10^{-35}$). The association remains negative at the neighbouring splits, with $\rho = -0.767$ for $7/8$ and -0.709 for $9/10$ (Fig. 2c; Supplementary Material 4). We further analyse the effective Weber fraction the model predicts, computing the coefficient of variation (CoV) of the fitted response distribution per numerosity and exposure (Fig. 2d). Model and data both settle to a roughly constant plateau at larger numerosities: a power-law fit over the width-based window $n \ge n_{\text{width}}$ gives a shallow pooled slope ($\gamma = -0.20$ for the model and -0.10 for the data), close to the constant coefficient of variation expected under Weber's law.

The sensory channel is phenomenological: s_0 sets the overall noise level of the initial estimate y_0, whereas λ_0 determines how that noise grows with numerosity. These parameters vary across exposure durations so that changes in input quality are not forced onto the shared landscape, and we do not assign them a further cognitive interpretation.

With these ingredients the model reproduces all three behavioural signatures together (Fig. 3): pooled across durations, $R^2 = 0.95$ for the mean estimate, 0.94 for exact accuracy, and 0.82 for response time, the last obtained from a constrained cross-channel fit in which the landscape was held fixed at its response-fit values and only the RT mapping was estimated from response times. There are two systematic deviations in the model predictions: the first is the 40-ms condition overall, which accumulates many deviations from the predictions, similarly to Cheyette and Piantadosi (2020). The other is a predicted shoulder around $n = 6$–9 in the accuracy data (Fig. 3b), and around $n = 5$–8 in the response time data (Fig. 3c), which are caused by the attractor dynamics (Supplementary Fig. S3), and not observed in the participants' behaviour.

Finally, we explored how the model would account for developmental dyscalculia by applying it to data on numerosity extraction of dyscalculic children versus age-matched controls from Decarli et al. (2020). The fitted landscapes of children with dyscalculia contained broader, more overlapping components and yielded a lower extrapolated width index (median $n_{\text{width}} = 8.1$ vs. 8.8; Mann–Whitney $p = .018$), consistent with a coarser fitted width profile (see Supplementary Material 5; Supplementary Fig. S4).

Discussion

We have described the inference of numerosity from noisy scalar evidence as pattern completion on an attractor

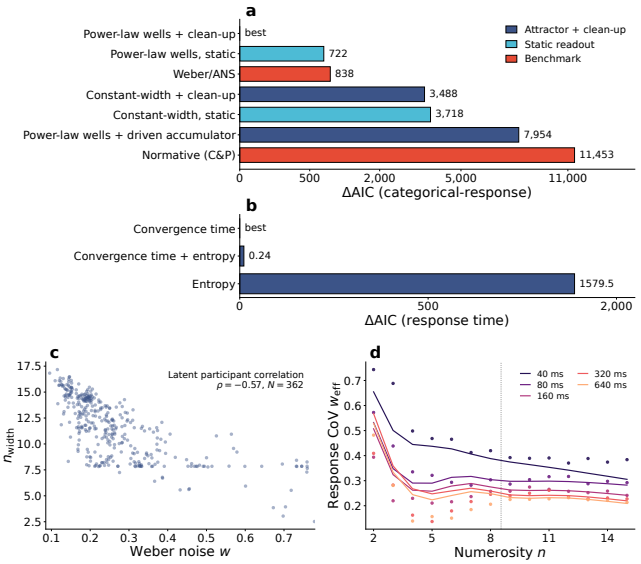


Figure 2. Model comparison and the link between landscape width and estimation acuity. a, Categorical-response channel: ΔAIC of each model relative to the winning power-law attractor with clean-up, over the $\sim$53,000 individual responses. Bars are coloured by model role: attractor with clean-up, matched static readout of the same wells, or external benchmark. Turning clean-up off, while retaining the same landscape and sensory parameterization but requiring no stopping-budget worsens the fit, and constant-width wells ($p = 0$, $\sigma_n = \sigma_c$) worsen it further. The purely driven attractor ("Power-law wells + driven accumulator", see Methods), whose input is the still-accumulating count rather than a completed estimate, fits far worse, because a partial count captures the wrong integer (Supplementary Material 2). Weber/ANS and the bounded-optimal model of Cheyette and Piantadosi (2020) ("Normative (C&P)") are shown for reference. **b**, Response-time channel: ΔAIC of the three RT links on the winning landscape, with one mapping shared across durations. Convergence time alone and the hybrid link perform almost identically, whereas entropy alone performs substantially worse. **c**, Individual differences estimated with a bivariate hierarchical model. Non-overlapping trials informed the two constructs: $n \leq 8$ for landscape resolution and $n \geq 9$ for Weber noise. Points show empirical-Bayes participant estimates on the original scale. The inset reports the hierarchical correlation ρ between the latent participant effects on $\log n_{\mathrm{width}}$ and $\log w$, after accounting for exposure duration and finite-trial uncertainty. The model controls for exposure duration and uses sessions with both 320- and 640-ms data. **d**, Effective Weber fraction: the coefficient of variation of the fitted response distribution per numerosity and exposure duration. Solid lines are the model, dots the data, and the dotted vertical marks the shared n_{width}; the coefficient of variation approaches a roughly constant plateau at larger numerosities.

landscape: the integers act like memories whose widths grow with magnitude, and an uncertain estimate relaxes towards the nearest resolvable region. Fitted to individual visual-enumeration responses, the model reproduces estimation, accuracy, and response time together and derives the loss of discrete resolution from the structure of the representation itself. The exact and approximate regimes become two readings of one numerical landscape rather than two systems joined at an unexplained seam.

The model has two main features that attempt to capture the cognitive mechanisms that define numerosity readout: the first is modelling the mental number line as an attractor landscape with integers placed in equally spaced Gaussian attractor wells that follow a power-law decline in resolution defined by numerical frequency. This suggests a possible link between the geometry of the representation and numerical experience (Anderson and Schooler 1991; Verguts and Fias 2004), so that frequent numbers are sharply encoded, rarer ones diffusely. Then, if indeed numerical experience shapes well width, development and training should progressively sharpen the fitted width profile. Across participants, a sharper fitted landscape already goes with lower estimation noise (larger n_{width}, smaller w; Fig. 2c). Over development, we predict that numerical experience should similarly alter the width profile. This change should be specific to exposure to *numbers*, not to general visual acuity: an intervention that sharpens the number code should differ from one that improves perception at large (Butterworth and Laurillard 2017). The developmental dyscalculia analysis (Supplementary Material 5) is an early step in this direction: the impaired group has a coarser fitted width profile than age-matched controls.

The second feature is the clean-up mechanism, which improves the fit substantially at the cost of a single parameter, the deliberation budget G (Methods). Furthermore, the attractor's convergence time predicts response time better than a static uncertainty proxy. The clean-up implements a Hopfield-style associative retrieval operation: each step moves the state towards a softmax-weighted average of the stored integers, so every well shapes the outcome. Because those patterns are the ordered points of the number line, not arbitrary memories, the softmax competition among neighbours is sharp for small, well-separated numbers and broad for large, overlapping ones. The graded loss of resolution and the lengthening of settling time towards the boundary both follow from this structure.

Nevertheless, this is strictly true only for the lower-numerosity response times, but without a bound, the predicted response time would keep rising while empirical RT plateaus (Fig. 1c). Response time therefore requires a bounded clean-up process for large numerosities. Without a processing limit, the dynamics slow markedly on the flat large-numerosity landscape. A single global clean-up budget G, determined from the categorical-response channel before fitting the RT link, converts this slowing into the observed large-number plateau. The plateau therefore does not follow from the energy landscape alone: it arises when the slowing clean-up dynamics interact with a finite stopping policy. The present model captures the behavioural consequence of that policy through G, while leaving its cognitive implementation open.

The model naturally accounts for the transition between the exact and approximate regimes of the core number system (Feigenson et al. 2004). As neighbouring wells broaden and overlap, clean-up becomes less effective and exact-readout probability falls gradually. The model's response distributions match the data across the whole range, and their coefficient of variation settles to a roughly constant plateau at larger numerosities, consistent with the scalar variability classically observed in non-verbal numerical estimation (Whalen et al. 1999; Cordes et al. 2001).

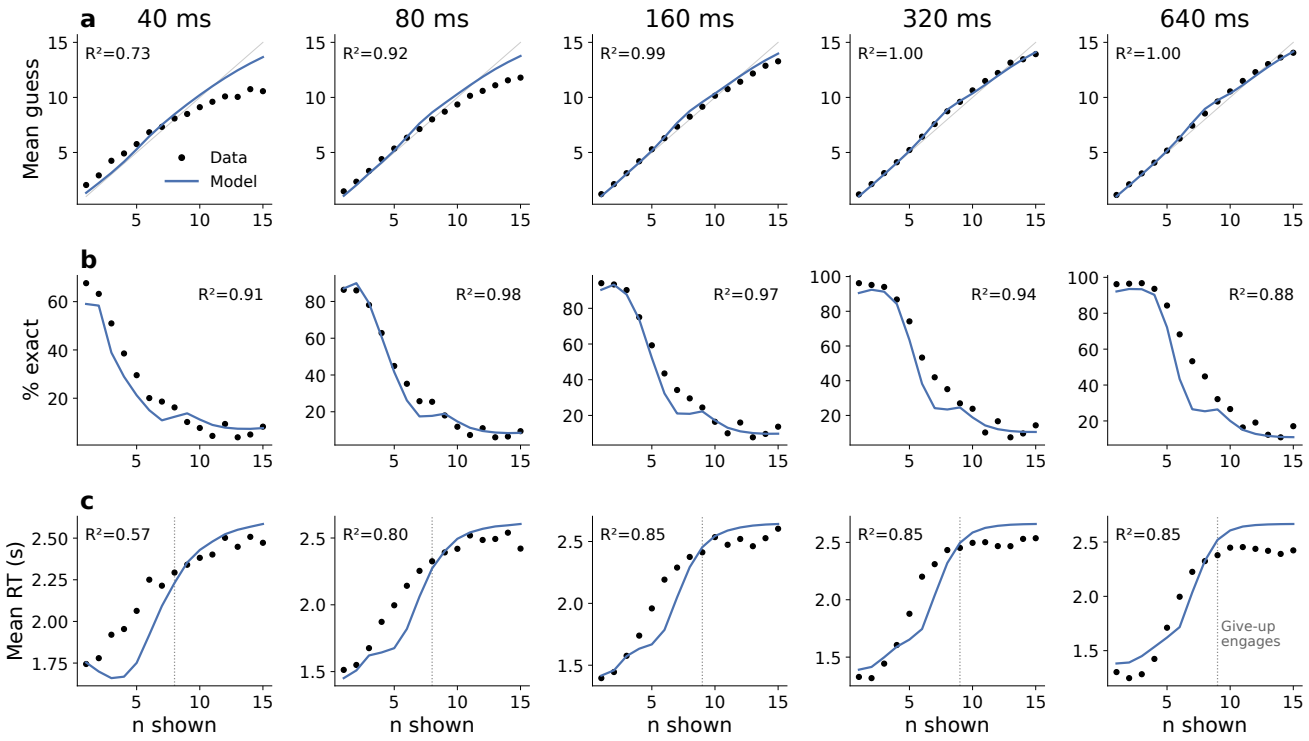


Figure 3. Fit of the joint model to the Cheyette and Piantadosi (2020) data across the three channels. Columns show presentation times from 40 to 640 ms; rows show the mean estimate, exact accuracy $P(r = n)$, and geometric-mean response time. Dots are empirical data; lines are the model, with the landscape fitted to the individual responses. Response time is evaluated in a constrained cross-channel fit: the landscape is held fixed at its response-fit values, and only the RT-link parameters are fitted to the individual trial times. The small shoulder in predicted exact accuracy occurs near the width-based proxy n_{width} and disappears when clean-up is switched off (Supplementary Fig. S3). R^2 values are given per panel.

There is nevertheless an important unexplained shoulder both in the accuracy and response time curves, generated by the integer clean-up dynamics but not clearly present in the behavioural data. This suggests that the transition may be smoother than in the fitted homogeneous landscape, potentially because of participant heterogeneity or variability in the stopping process.

The attractor and visuospatial-capacity accounts are competing explanations of where the same psychophysical regularities arise. A shared numerical landscape predicts that its geometry should generalize across sensory modalities, with modality-specific sensory channels feeding the same number representations; cross-modal numerosity adaptation provides an empirical precedent for this possibility (Arrighi et al. 2014). If the landscape is plastic, its resolution should covary with numerical experience and may be sharpened by numerical learning, as suggested by associations between nonsymbolic acuity and mathematics achievement and by comparisons across levels of schooling (Halberda et al. 2008; Piazza et al. 2013; Nys et al. 2013). However, evidence that nonsymbolic acuity can remain robust with little formal schooling (Pica et al. 2004; Butterworth et al. 2008; Frank et al. 2008; Zebian and Ansari 2012) constrains a strongly schooling-dependent interpretation of the landscape. Formal numerical education may sharpen the representation, but cannot be its sole source; nonsymbolic experience with quantity must also contribute, or part of the landscape must emerge independently of training. Finally, atypical landscape geometry could also contribute to some forms of developmental dyscalculia (Piazza et al. 2010; Mazzocco et al. 2011; Decarli et al. 2023; Dolfi et al. 2024).

The individual-differences result further connects the two behavioural regimes. Landscape resolution estimated from the lower-numerosity responses predicts Weber noise estimated from a non-overlapping set of larger-numerosity trials, even after accounting for exposure duration and estimation uncertainty. This cross-range association is consistent with one individual representational geometry constraining both categorical and approximate behaviour, rather than n_{width} merely redescribing the dispersion of the same large-number responses. The association is correlational, but it provides participant-level construct validity for the width profile.

A visuospatial-capacity account instead predicts shared individual limits across non-numerical visual-memory and numerosity tasks. Consistent with this prediction, Cheyette et al. (2024) found that information-capacity estimates from a non-numerical change-localization task generalized to number estimation within participants. A related visual proposal locates numerosity in the recurrent dynamics of an attentional-priority (saliency) map, where subitizing and estimation emerge from one network as lateral inhibition varies, without a separate mechanism for each range (Sengupta et al. 2014, 2017; Verma and Sengupta 2023). This account shares the present model's recurrent, attractor-like dynamics, which shows that the dynamics themselves do not separate the two proposals; what differs is the locus of the code. There, numerosity is a summary statistic read from a modality-specific map of visual

space; here, it is a theoretically amodal representation on the number line that such sensory maps would feed. The accounts therefore make contrasting predictions: cross-modal parameter generalization and selective effects of numerical training would favour a shared numerical landscape, whereas transfer from general visuospatial-capacity interventions would favour the visual-capacity account.

The current model was designed to investigate a putative amodal parietal code supporting numerical inference and therefore deliberately omits a sensory front end. Stoianov and Zorzi (2012) showed that numerosity-selective representations can emerge from unlabelled pixel-level visual input through unsupervised learning. Including a visual front end here would have conflated the sensory extraction of numerosity with the amodal numerical computations that the model was designed to isolate. This fact also implies that this account is a purely numerical readout and does not model spatial or attentional manipulations that may contribute to classical subitizing effects. Future work towards a complete perceptual account of numerosity extraction would start by including a sensory front end that could account for these factors, among others. One option would be to couple the landscape to an image-computable front end, such as those developed by Zorzi and colleagues (Testolin et al. 2020; Dolfi et al. 2024) or by Sengupta and colleagues (Verma and Sengupta 2023). Such a model would connect sensory encoding to attractor dynamics. Analogous extensions to auditory and tactile numerosity would provide a direct test of cross-modal parameter sharing.

The failure of the purely driven accumulator clarifies the division of labour between numerical accumulation and associative retrieval. A partial count is not a degraded version of the final cardinality: every prefix is itself a valid numerosity and remains compatible with any larger total until the set or sequence is complete. A cardinality attractor therefore cannot infer the final number from the running count alone. In the present architecture, modality-specific processes first construct an estimate of the completed set or sequence, after which the shared landscape stabilizes that estimate. This still permits serial accumulation, but places pattern completion after the relevant evidence has been delimited rather than within the counting process itself. Finally, the present analysis tests the simplest power-law width schedule motivated by numerical frequency. Whether alternative smooth width functions provide a better description, and how such alternatives would affect the remaining model parameters, remain questions for future work.

Numerical readout, in this framework, is pattern completion on a landscape of number memories. Within the numerical readout, more categorical and approximate reports become two readings of one landscape rather than two codes joined at an unexplained seam. The bounded convergence time of the clean-up dynamics also captures the main response-time profile, although the cognitive implementation of the stopping policy remains open. The result is a compact, mechanistic set of hypotheses about how the mind represents number.

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Author Contributions

Eudald Correig-Fraga: Conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing - original draft, and writing - review and editing.

Ethical Considerations

This article reports computational modelling and secondary analysis of previously published, publicly available datasets. No new human participants were recruited, and no new human data were collected for this study. Ethical approval for the original data collections was reported in the source publications.

Consent to Participate

Not applicable. This study did not recruit participants or collect new participant data.

Consent for Publication

Not applicable. This study does not contain identifiable individual participant information.

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Declaration of conflicting interests

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Data Availability

Code is available at https://github.com/eudaldc/hopfield_numbers. The source datasets are available at https://osf.io/svcy5/ and https://osf.io/gjpb3/.

Supplemental material

Supplementary material is available with this article.

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